

Abstract:

This project focuses on developing an AI-driven image generator website using the MERN stack (MongoDB, Express.js, React, Node.js) for seamless integration of frontend and backend technologies. The platform incorporates advanced deep learning models like Stable Diffusion, to generate high-quality and customizable images based on user-provided textual prompts.

The application features an intuitive user interface built with React, enabling users to input prompts, and view generated images in real-time. The backend, powered by Node.js and Express, manages requests and orchestrates the AI model, while MongoDB stores user preferences and generated images for a personalized experience. Optimization techniques ensure efficient performance, even within constrained computational environments.

This project is designed to cater to the creative needs of artists, designers, and enthusiasts, offering a scalable, user-friendly, and robust platform for generating custom digital artwork. The integration of cutting-edge AI models with a modern web stack ensures a powerful and accessible solution for the next generation of creative tools